



Dropout Universality

Critical Scaling and Optimal Scheduling at the Edge-of-Chaos

Lucas Fernandez-Sarmiento

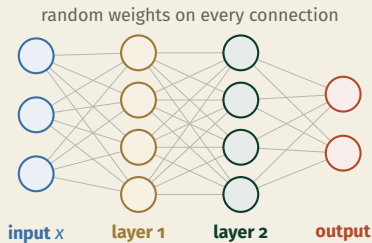
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Usual view of Multi-Layer Perceptrons



Circles represent neurones, edges are interactions.

One layer: affine map, then nonlinearity ϕ

$$\mathbf{a}^{(0)} = \mathbf{x}, \quad \mathbf{z}^{(\ell)} = W^{(\ell)} \mathbf{a}^{(\ell-1)} + \mathbf{b}^{(\ell)}$$

$$\mathbf{a}^{(\ell)} = \phi(\mathbf{z}^{(\ell)})$$

ϕ controls how strongly a weighted input activates a neuron.

At initialization

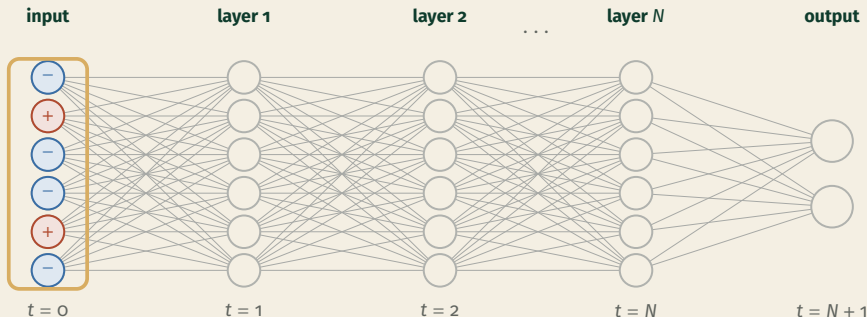
$$W_{ij}^{(\ell)} \stackrel{\text{iid}}{\sim} \mathcal{N}\left(0, \frac{\sigma_w^2}{n_{\ell-1}}\right), \quad b_i^{(\ell)} \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_b^2),$$

$$\mathbf{x} \sim p_{\text{data}}.$$

Weights are usually zero-mean Gaussians, with variance scaled by the width of the preceding layer, $n_{\ell-1}$.

At initialization, we just have some random number generator

Toy task: decide whether the input contains more (+) or more (-) signs.



One can see layer updates as some RG coarse graining flow equation.

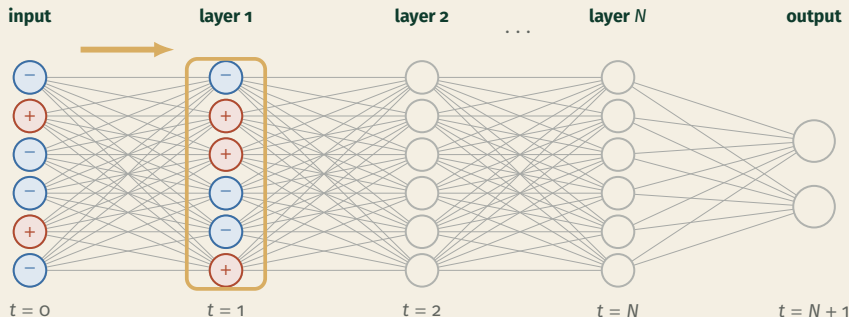
$$\mathbf{s}^{(\ell+1)} = \text{sign}\left(W^{(\ell+1)}\mathbf{s}^{(\ell)} + \mathbf{b}^{(\ell+1)}\right), \quad s_i \in \{-1, +1\}.$$

+ and - show the sign of each activation.

The input has a negative majority, but random weights do not yet preserve it.

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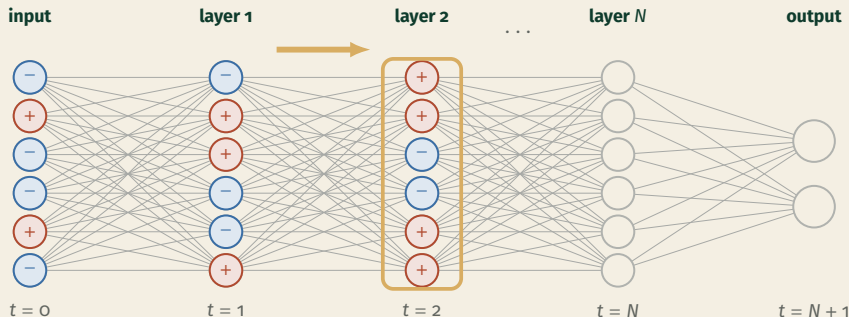
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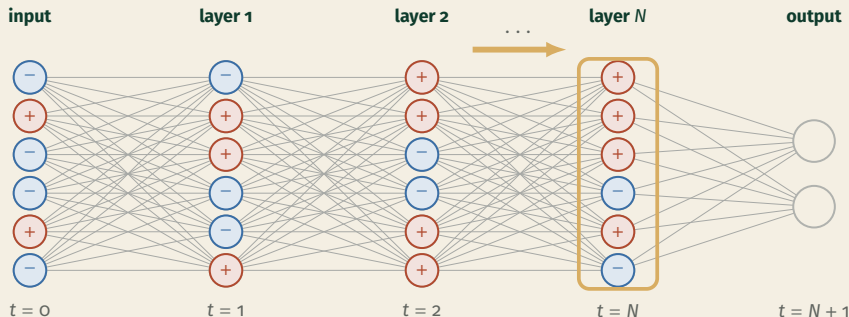
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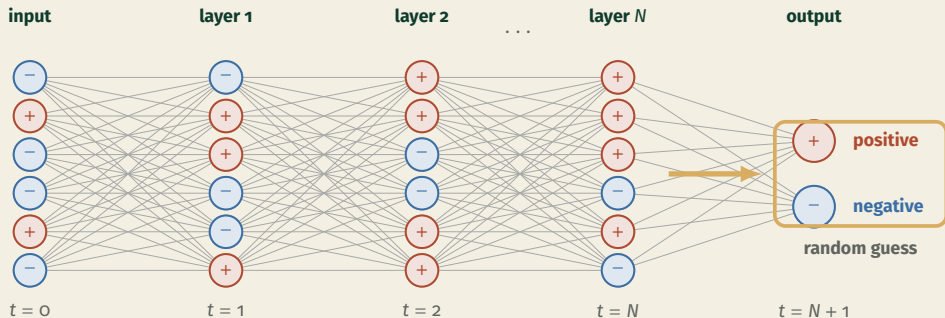
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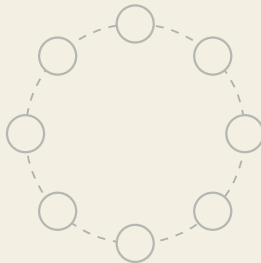
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A dense layer as an all-to-all spin system

one layer's state



rearrange a layer into a circle



A dense layer as an all-to-all spin system

Neural-network variables

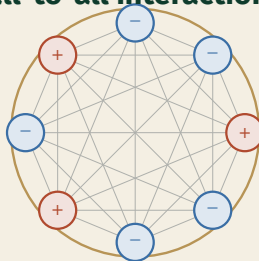
activation $a_i \leftrightarrow$ spin s_i

weight $W_{ij} \leftrightarrow$ coupling J_{ij}

bias / dropout $\leftrightarrow$ external field



all-to-all interactions



Ising ferromagnet

nearest-neighbour,
uniform

$$J_{ij} = J \text{ on } \langle ij \rangle$$

all-to-all

Curie–Weiss

all-to-all, uniform
 $J_{ij} = J/N$ for every pair

quenched disorder

Edwards– Anderson

nearest-neighbour,
quenched random
random J_{ij} on $\langle ij \rangle$

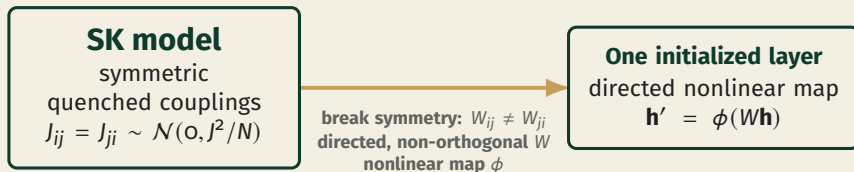
all-to-all

quenched disorder

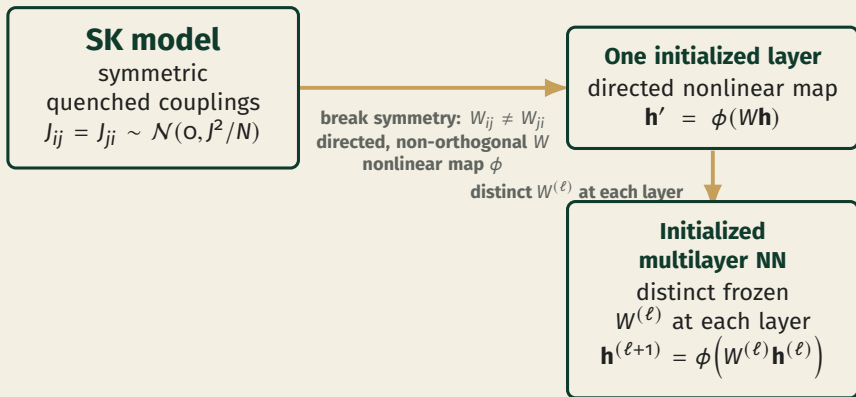
Sherrington– Kirkpatrick

all-to-all, quenched
random symmetric
 $J_{ij} = J_{ji} \sim \mathcal{N}(0, J^2/N)$

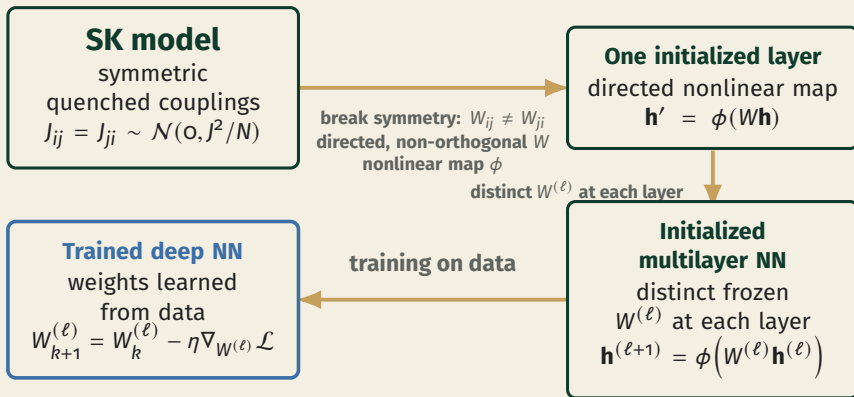
From SK-like disorder to a trained dense network



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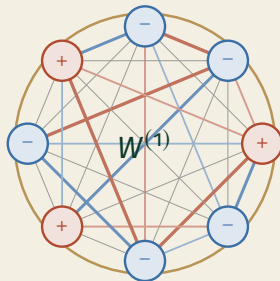
Before training, random couplings scramble the majority

**Before training:
a new random matrix**

$W^{(1)}, W^{(2)}, \dots, W^{(N)}$

scramble the
interactions

layer 1 $\longleftrightarrow$ time $t = 1$



Coupling key

— $W_{ij} > 0$

— $W_{ij} < 0$

— small $|W_{ij}|$

configuration $\mathbf{s}^{(1)}$

Outputs are essentially random

$$\mathbf{s}^{(\ell+1)} = \text{sign}\left(W^{(\ell+1)}\mathbf{s}^{(\ell)} + \mathbf{b}^{(\ell+1)}\right).$$

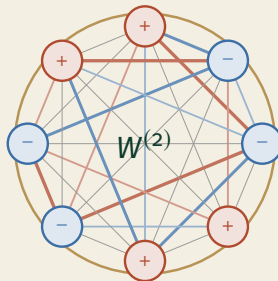
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scramble the
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layer 2 $\longleftrightarrow$ time $t = 2$



configuration $\mathbf{s}^{(2)}$

Coupling key

— $W_{ij} > 0$

— $W_{ij} < 0$

— small $|W_{ij}|$

Outputs are essentially random

$$\mathbf{s}^{(\ell+1)} = \text{sign}\left(W^{(\ell+1)}\mathbf{s}^{(\ell)} + \mathbf{b}^{(\ell+1)}\right).$$

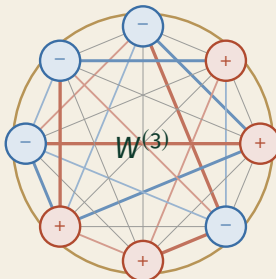
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scramble the
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layer 3 $\longleftrightarrow$ time $t = 3$



Coupling key

- $W_{ij} > 0$
- $W_{ij} < 0$
- small $|W_{ij}|$

configuration $\mathbf{s}^{(3)}$

Outputs are essentially random

$$\mathbf{s}^{(\ell+1)} = \text{sign}\left(W^{(\ell+1)}\mathbf{s}^{(\ell)} + \mathbf{b}^{(\ell+1)}\right).$$

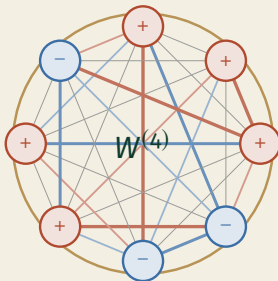
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**Before training:
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$W^{(1)}, W^{(2)}, \dots, W^{(N)}$

scramble the
interactions

layer 4 $\longleftrightarrow$ time $t = 4$



configuration $\mathbf{s}^{(4)}$

Coupling key

- $W_{ij} > 0$
- $W_{ij} < 0$
- small $|W_{ij}|$

Outputs are essentially random

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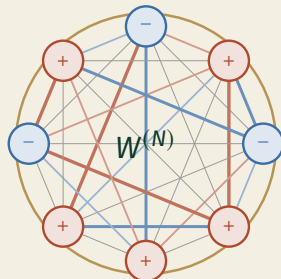
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**Before training:
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scramble the
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layer $N \longleftrightarrow$ time $t = N$



Coupling key

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configuration $\mathbf{s}^{(N)}$

Outputs are essentially random

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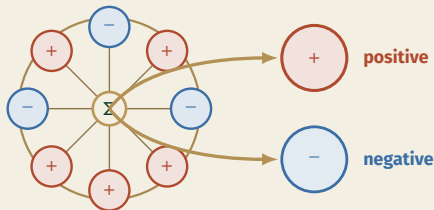
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final configuration $\rightarrow$ class scores



**No learned
coarse-graining**

The negative majority
has been lost.

pool all eight activations, then score each class

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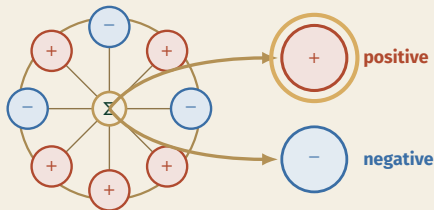
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a confident but random guess

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With training, the network coarse-grains the majority

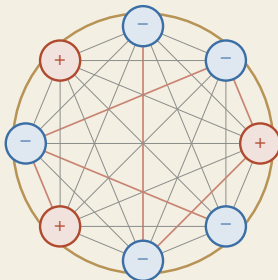
A slight negative majority becomes a stable, increasingly ordered signal.

**Five negative,
three positive**

Each trained layer
removes ambiguity

Like coarse-graining:
local detail disappears,
the majority survives.

layer 0 $\longleftrightarrow$ time $t = 0$



**The learned couplings
favor alignment**

— stronger
— weaker

configuration $s^{(0)}$

Training turns depth into a useful flow

5- : 3+ $\longrightarrow$ 6- : 2+ $\longrightarrow$ 7- : 1+ $\longrightarrow$ 8- : 0+ .

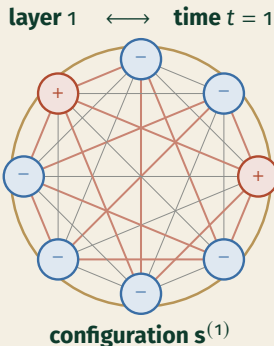
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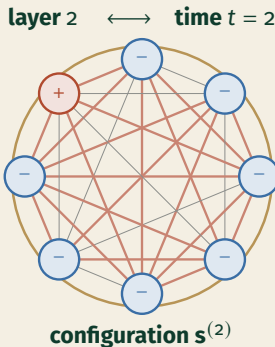
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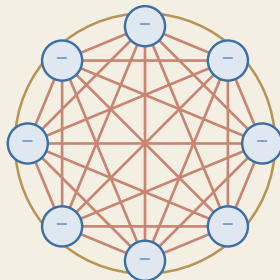
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configuration $s^{(N)}$

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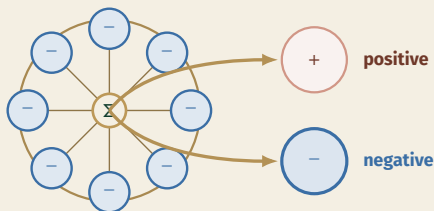
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pool all activations, then sample from outputs.

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5- : 3+ → 6- : 2+ → 7- : 1+ → 8- : 0+ .

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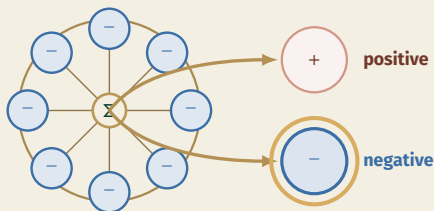
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negative class wins

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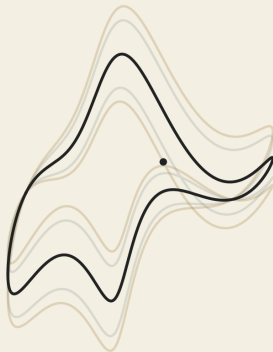
5- : 3+ → 6- : 2+ → 7- : 1+ → 8- : 0+ .

In search of better variables

*“With four parameters I can fit an elephant,
and with five I can make him wiggle his
trunk.”*

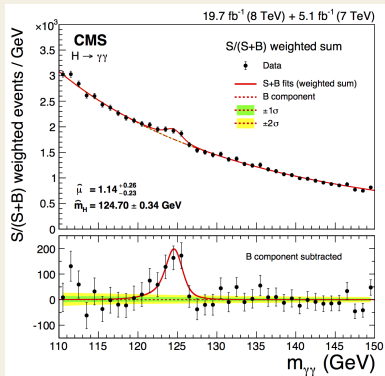
von Neumann to Fermi, via Dyson

Can a few **collective variables** describe
what depth does to information?



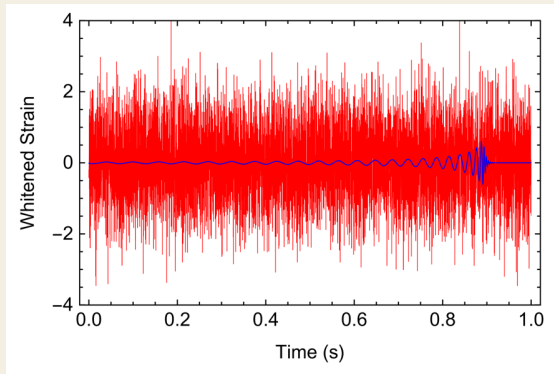
Mayer, Khairy & Howard,
Am. J. Phys. **78**, 648 (2010)

Finding signals in noise is hard!



Higgs signal at CERN

A small excess above a large background



Gravitational-wave signal

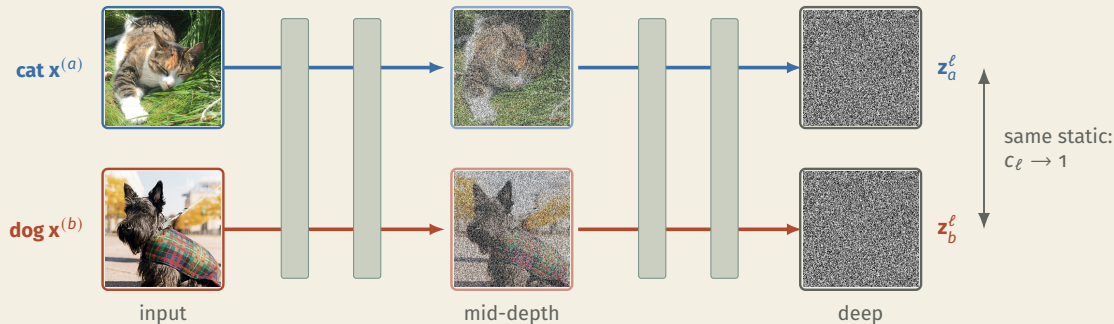
A structured waveform buried in detector noise

The structure we care about can be much smaller than the fluctuations around it.

A neural broken telephone

Each layer *re-transmits* the message it receives. Track the correlation between the two inputs, c_ℓ .

Aim of the game: postpone perfect correlation for as long as possible!



Living on the edge (of chaos)

Ordered

$c_\ell \rightarrow 1$ fast;
everything looks alike

ordered



Critical edge

forgetting is slow;
distinctions survive deep

edge-of-chaos



Chaotic

differences blow up,
then saturate

chaotic



layer 0

A circular input manifold through random tanh layers, projected to 2D.

Living on the edge (of chaos)

Ordered

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ordered



Critical edge

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edge-of-chaos



layer 10

Chaotic

differences blow up,
then saturate

chaotic



A circular input manifold through random tanh layers, projected to 2D.

Living on the edge (of chaos)

Ordered

$c_\ell \rightarrow 1$ fast;
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Critical edge

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layer 40

Chaotic

differences blow up,
then saturate

chaotic



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Living on the edge (of chaos)

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layer 100

Chaotic

differences blow up,
then saturate

chaotic

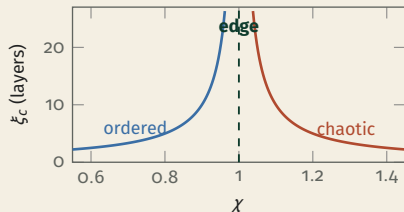


A circular input manifold through random tanh layers, projected to 2D.

Tune initialization to the edge so distinctions propagate deepest.

Criticality buys trainable depth

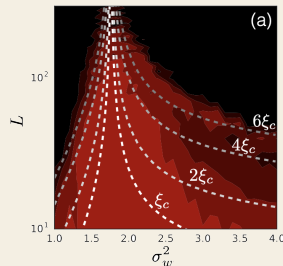
Criticality's signature: diverging correlation lengths



Criticality means a **diverging correlation length**,
 $\xi_c \rightarrow \infty$, counted here in *layers*.

Many systems work best at the "edge-of-chaos".

Free lunch?

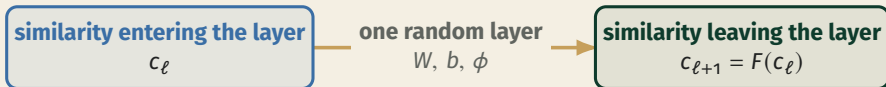


Training accuracy after a fixed number of updates, over σ_w^2 and depth L . In the referenced MLP experiment, **the trainable boundary approximately tracked multiples of the correlation length.**

Schoenholz et al. 2017.

ξ_c sets an effective useful depth for NNS

At initialization, one layer induces a map of similarity



What is F ?

At large width, the CLT makes the two preactivation fields u, v jointly Gaussian. The layer map is effectively

$$F(c) \sim \int \mathcal{D}_c(u, v) \phi(u) \phi(v) \sim \langle V(u) V(v) \rangle_c.$$

a vertex-insertion correlator

Repeating layers means iterating F

$$c_0 \xrightarrow{F} c_1 \xrightarrow{F} c_2 \xrightarrow{F} \cdots \longrightarrow c^*, \quad F(c^*) = c^*.$$

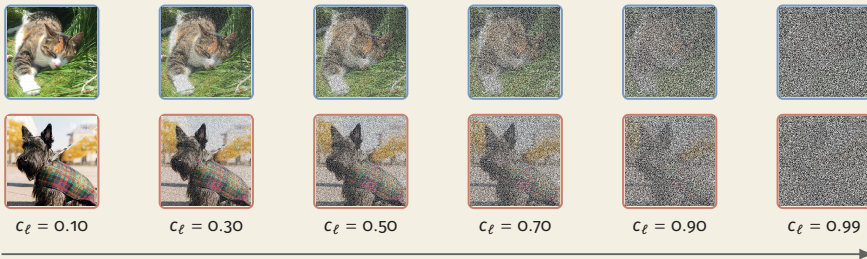
Near the fixed point,

$$c_{\ell+1} - c^* \approx F'(c^*)(c_\ell - c^*).$$

How fast does the correlation settle onto its fixed point c^* ? Linearize the map about it.

$$|c_\ell - c^*| \sim |F'(c^*)|^\ell = e^{-\ell/\xi_c}, \quad \xi_c^{-1} = -\log |F'(c^*)| = -\log \left[\sigma_w^2 \int Dz_1 Dz_2 \phi'(u_1^*) \phi'(u_2^*) \right].$$

σ_w the weight std, ϕ the nonlinearity, $u_{1,2}^*$ the preactivations at the fixed point. So ξ_c is set by σ_w, σ_b, ϕ alone; we tune those to get criticality. Below, at the edge and without dropout, $c^* = 1$.



Tune to the edge and $\xi_c \rightarrow \infty$: collapse becomes subexponential.

Learning should survive missing information

Proof

Let $u, v \in V$ over $\mathbb{R}$. For any scalar $t \in \mathbb{R}$, the squared norm is non-negative:

$$\|u + tv\|^2 \geq 0$$

Expanding via the inner product:

$$\langle u + tv, u + tv \rangle = \|u\|^2 + 2t\langle u, v \rangle + t^2\|v\|^2 \geq 0$$

Since this quadratic polynomial in t is non-negative for all real t , its discriminant must be non-positive ($\Delta \leq 0$):

$$(2\langle u, v \rangle)^2 - 4\|u\|^2\|v\|^2 \leq 0$$

Dividing by 4 and rearranging:

$$\langle u, v \rangle^2 \leq \|u\|^2\|v\|^2$$

Taking the principal square root yields the result:

$$|\langle u, v \rangle| \leq \|u\|\|v\| \quad \blacksquare$$

Complete proof

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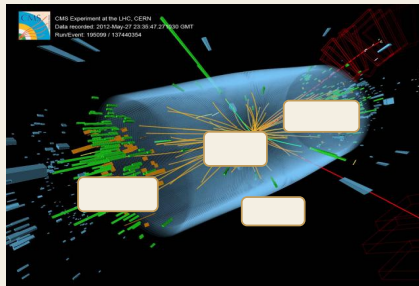
Infer the missing step

A student (or LLM) should extrapolate even when parts of the training signal are missing.



Gravitational lensing

Drop a patch and a repeated galaxy or star vanishes

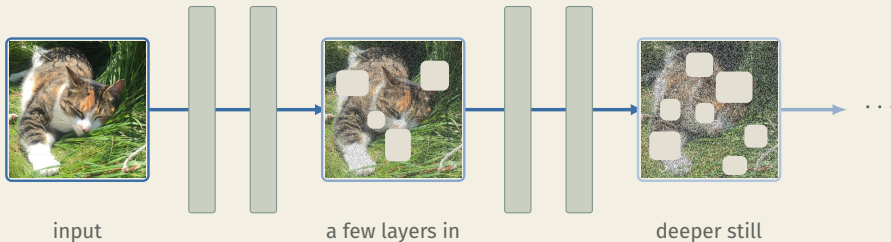


CERN collision event

Drop a patch and part of a jet disappears

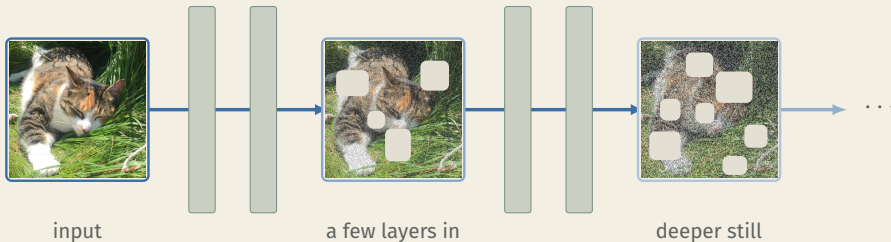
We want to detect events even when part of the signal is corrupted—or an instrument partially fails.

A usual suspect of regularization: dropout



A fresh mask at every layer: the pattern changes and the damage compounds.

A usual suspect of regularization: dropout



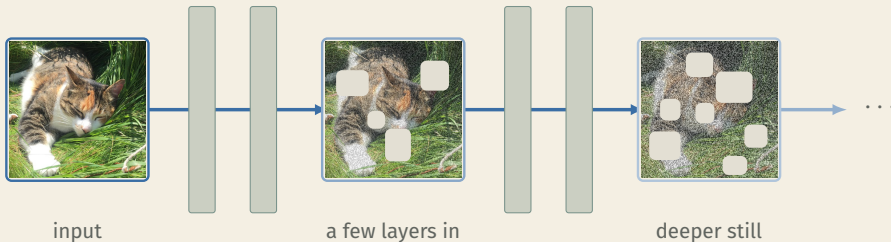
A fresh mask at every layer: the pattern changes and the damage compounds.

The standard theory

Usual NNs are overparametrized, so there are enough parameters to memorize training neural patterns ("coadaptation"). Dropout blocks activations randomly, mitigating this.

Srivastava et al., JMLR **15**, 1929 (2014).

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A fresh mask at every layer: the pattern changes and the damage compounds.

The standard theory

Usual NNs are overparametrized, so there are enough parameters to memorize training neural patterns ("coadaptation"). Dropout blocks activations randomly, mitigating this.

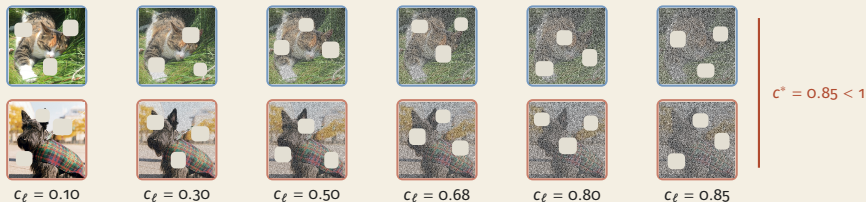
Srivastava et al., JMLR **15**, 1929 (2014).

The practice

People usually do a grid search over the probability of dropping out and call it a day.

Reasonable. But is it **optimal**?
Can we do better?

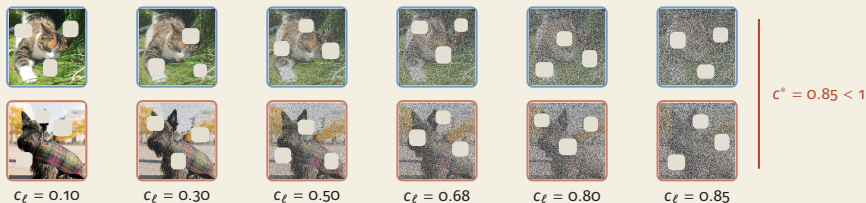
Old problems... old solutions?



A fresh, independent mask is drawn for each input at every layer, so even in pure noise the two never agree: c_ℓ stalls below 1.

$c = 1$ stops being a fixed point, and ξ_c was a linearization *about* it, so it is not even defined! Look for the fixed point *perturbatively* away from perfect correlation, in the dropout field $h \propto 1 - \rho$.

Old problems... old solutions?



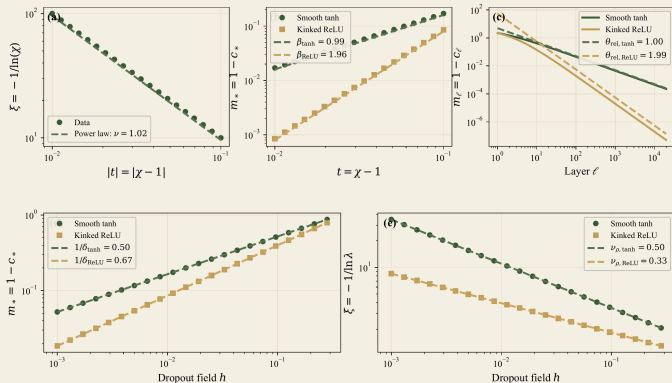
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Tuned to the edge ($t \equiv \chi_\rho - 1 = 0$), one can still define an effective correlation length:

$$\xi_c \propto h^{-1/2}.$$

Universality is only sensitive to analyticity



Critical exponents for different networks

	smooth	kinked	Ising
$\xi \sim t ^{-\nu_t}$	1	1	$\frac{1}{2}$
$m \sim t ^\beta$	1	2	$\frac{1}{2}$
$m_\ell \sim \ell^{-\theta}$	1	2	1
$m \sim h^{1/\delta}$	2	$\frac{3}{2}$	3
$\xi \sim h^{-\nu_\rho}$	$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{3}$

The statmech model of dropout MLPs

- t $t \equiv \chi_\rho - 1$: how far the network sits **from criticality**, with $\chi_\rho = \bar{F}'_\rho(1)$.
the reduced temperature. $t = 0$ is the edge of chaos.
- m $m \equiv 1 - c^*$: how far the fixed point sits **from perfect correlation**.
the order parameter, i.e. the magnetization. $m = 0$ is the ordered phase.
- h $h \equiv 1 - \bar{F}_\rho(1) \propto 1 - \rho$: **dropout**.
an external field conjugate to m , which biases the network away from alignment whether or not it is critical.

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Landau equation of state

$$h = \frac{g_\rho}{2} m^2 - t m$$

$$m(t, h) = \frac{t + \sqrt{t^2 + 2g_\rho h}}{g_\rho}$$

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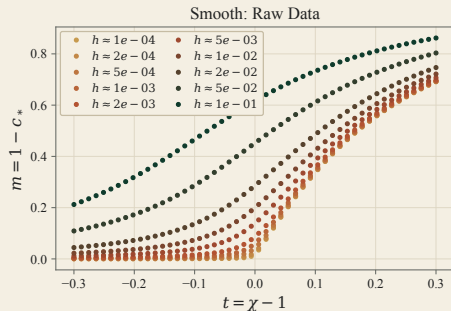
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The right variables give us a regime separation and a scaling collapse.

$$\tilde{m} \equiv m \sqrt{\frac{g_\rho}{2h}}, \quad \tilde{t} \equiv -\frac{t}{\sqrt{2g_\rho h}}$$
$$\tilde{m} = \sqrt{1 + \tilde{t}^2} - \tilde{t}$$

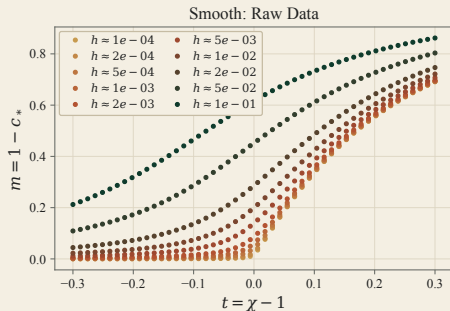
Behold universality: granular details such as σ_w , ϕ , cease to matter: every smooth network, at every detuning and every dropout rate, observe the same behaviour, and detuning and dropout reveal themselves as two sides of the same coin.

Scaling collapse

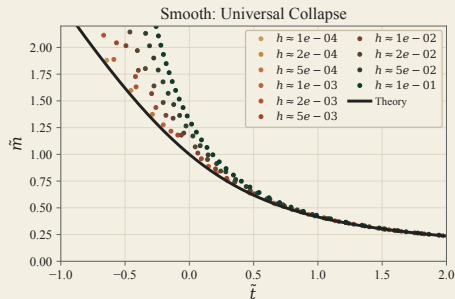


Sweep dropout strength h .

Scaling collapse



Sweep dropout strength h .



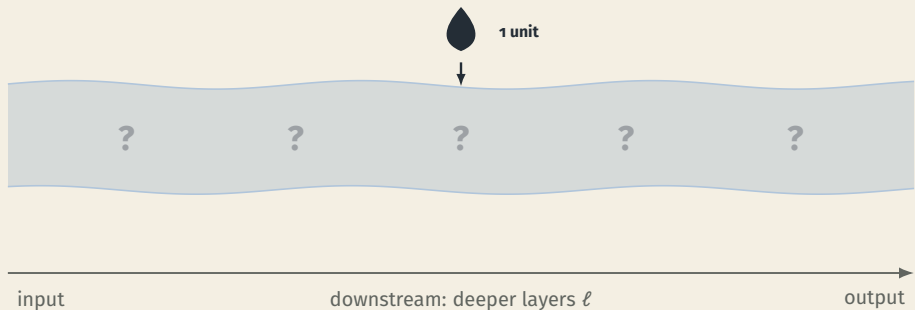
Rescale by $\tilde{m} = m\sqrt{g_\rho/2h}$, $\tilde{t} = -t/\sqrt{2g_\rho h}$, and the collapse becomes evident.

The black line is the exact prediction $\tilde{m} = \sqrt{1 + \tilde{t}^2} - \tilde{t}$.

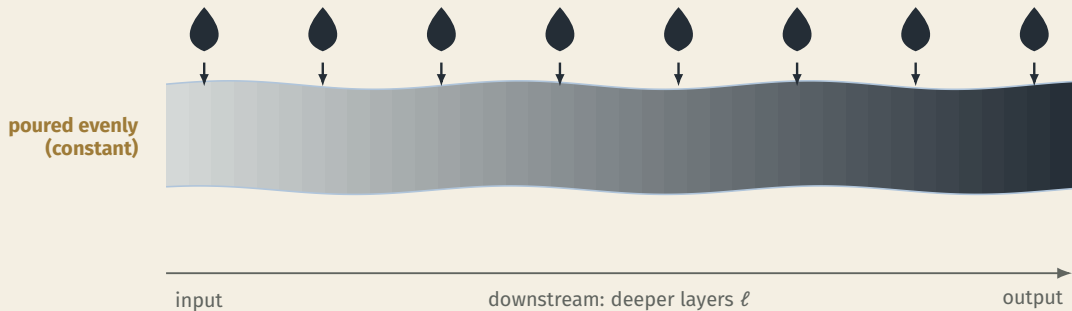
In QFT language, Callan–Symanzik at a fixed point: $\beta(g^*) = 0$, correlators go homogeneous, stripping the power leaves one scaling function.

Intuition: where would you pour the ink?

Say you have one unit of ink, and you want to corrupt a river as much as possible. Where would you pour it?

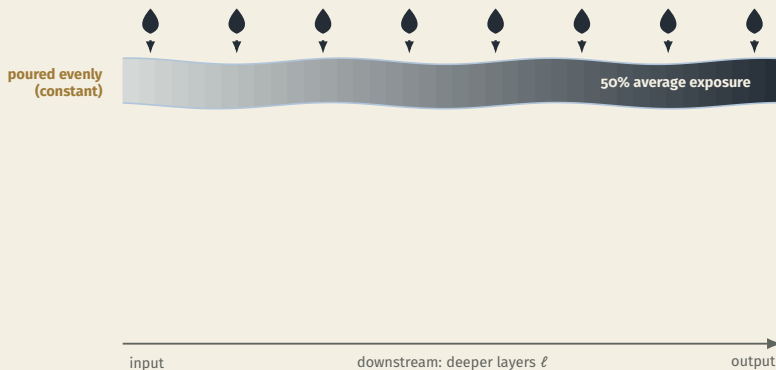


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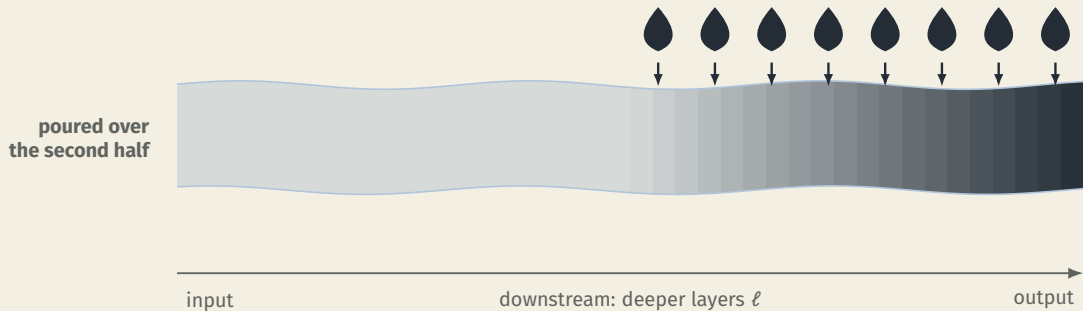


Rate $1/L$ on every layer: the water darkens the whole way down, and only reaches full strength as it leaves.

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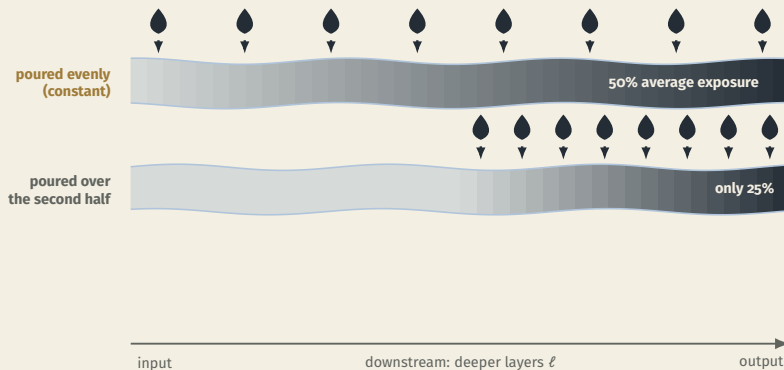


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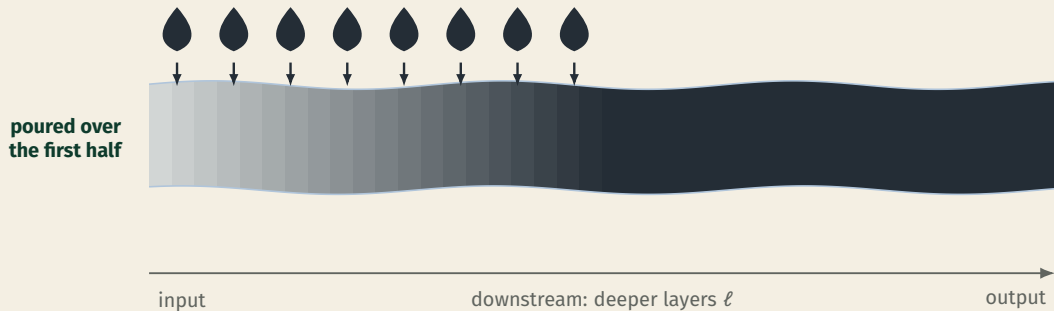


Rate $2/L$ darkens the water twice as fast, but it starts halfway down and is gone almost at once.

Intuition: where would you pour the ink?

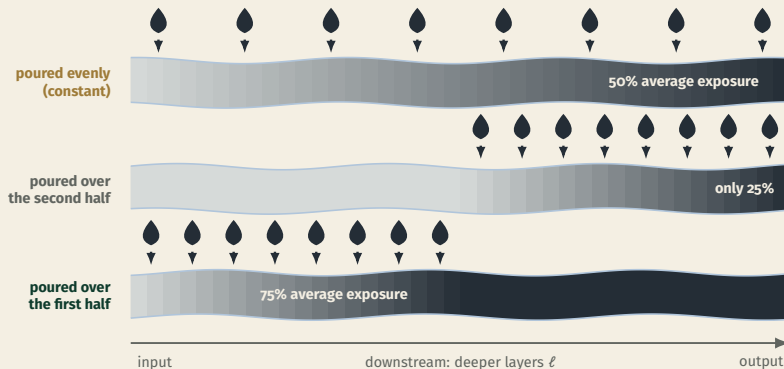


Intuition: where would you pour the ink?



The same $2/L$ ramp, but finished by the midpoint, so the entire second half runs at full strength.

Intuition: where would you pour the ink?



We should clearly drop the ink upstream for maximal corruption given a set amount of ink.

A more rigorous approach

Depth acts like an RG scale: the couplings *run*, so it is natural to treat them as functionals in depth, $h \rightarrow h_\ell$.

Then make ξ_{eff} a functional of the schedule and maximize it at fixed budget:

$$\max_{\{h_\ell\}} \xi_{\text{eff}}[h] \quad \text{s.t.} \quad \frac{1}{L} \sum_{\ell} h_\ell = \bar{h}, \quad 0 \leq h_\ell \leq h_{\text{max}}.$$

constant



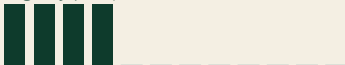
linear front



step front, $f = 1/2$



big step, $f = 1/3$



layer ℓ

Four profiles, identical $\sum_{\ell} h_\ell$.

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At the edge $\xi_{\text{eff}}^{-1}[h] \propto \frac{1}{L} \sum_{\ell} h_\ell^a$ with $a < 1$, so maximizing ξ_{eff} means minimizing a **concave** sum at fixed mean. The optimum sits at the corners: $h_\ell \in \{0, h_{\text{max}}\}$.

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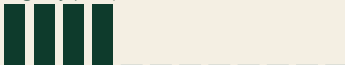
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Mean field theory gives us the correlation length. It is order-invariant, so it cannot say *where* the block goes.

Reach fixes the position. Later representations benefit from earlier masks too.

constant



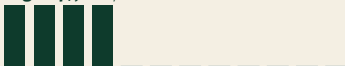
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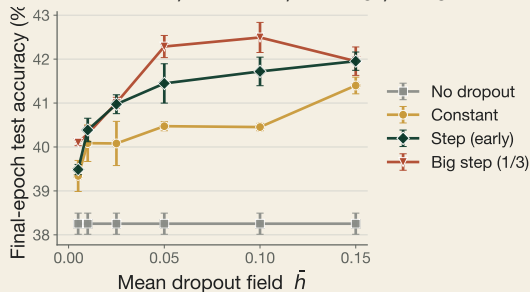
layer ℓ

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The predicted schedule wins, and wins sooner

Accuracy across the budget sweep

CIFAR-10, ReLU MLP, width 256, $n = 3$

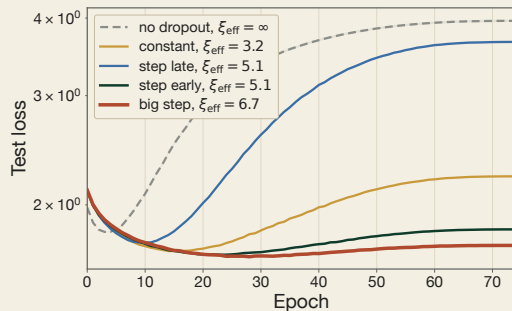


Mean $\pm$ s.e. over independent runs; both panels are final-epoch test performance.

The harder you front-load at fixed budget, the longer ξ_{eff} and the flatter the curve. Step early and step late share $\xi_{\text{eff}} = 5.1$ and still separate: that gap is reach, not propagation.

Test loss, same budget $\bar{h}$

depth $L = 6$, $\bar{h} = 0.1$, $n = 25$



Final-epoch test loss

18–35% lower

MLP settings

4–6% lower

Vision Transformers

Epochs to match constant dropout

29–31 VS 75

≈60% fewer, at equal per-epoch cost

“Fewer epochs” is epoch-proportional compute at identical architecture and batch size, not wall-clock or per-step FLOPs.

Across benchmarks, scheduling helps in 9 of 13 settings

Dataset / model	Dropout vs none	Schedule vs uniform	Scheduling
	acc. ↑ / loss ↓	acc. ↑ / loss ↓	
MLPs			
Amazon 2k	+9.02 pp / 7.79%	+7.86 pp / 4.26%	strong
Amazon 5k	+2.46 pp / 9.55%	+4.04 pp / 7.54%	strong
Jannis	+3.44 pp / 3.22%	+2.04 pp / 3.25%	strong
Speech	+6.91 pp / 22.15%	+1.84 pp / 12.71%	strong
Tiny ImageNet 20k	+0.96 pp / 1.70%	+1.40 pp / 1.94%	clear
Amazon original	+0.90 pp / 2.63%	+1.21 pp / 5.12%	clear
Amazon 20k [†]	+0.98 pp / 2.29%	+0.80 pp / 3.58%	clear
Transformers			
Speech	+3.62 pp / 12.44%	+1.73 pp / 4.91%	strong
Tiny ImageNet 20k	+1.35 pp / 2.30%	+1.63 pp / 2.17%	clear
Tiny ImageNet 80k	+2.20 pp / 3.88%	+0.29 pp / 1.01%	marginal
Amazon original	+0.42 pp / 6.40%	0 pp / 3.40%	loss only
Amazon 5k	+2.36 pp / 5.55%	0 pp / 0.15%	late only
Amazon 20k	+0.92 pp / 7.85%	0 pp / 0%	none

Five-seed means. A meaningful benefit is ≥ 0.9 pp accuracy or $\geq 2\%$ lower best-checkpoint test loss.

[†]Two confirmations remain missing from non-winning Amazon-20k MLP profiles.

All seven MLP settings benefit from scheduling; Transformer gains are dataset-dependent.

From microscopic masks to a macroscopic design rule

Dropout hurts propagation.



Maximize the correlation length.

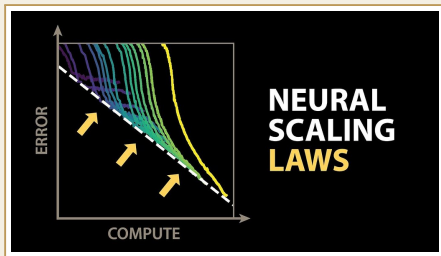


Front-load dropout.

Outlook: treat other depth-dependent hyperparameters as dynamical fields.

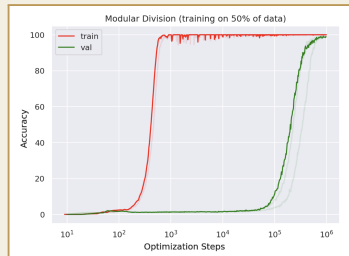
Questions?

arXiv:2605.21648 · Carnegie Mellon University



A theory of neural scaling laws

Derive the exponents and crossover scales from the data, architecture and optimizer—not only from empirical fits.

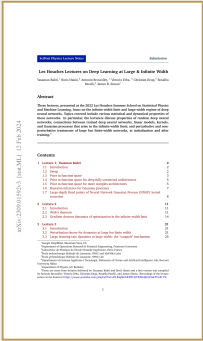


A physical picture of grokking

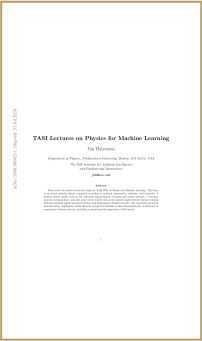
Explain delayed generalization as a dynamical transition in representation space.

Can one theory predict both smooth power laws and sudden learning transitions?

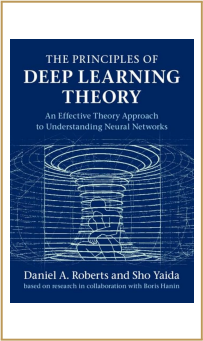
Recommended readings



Yasaman Bahri et al.
arXiv:2309.01592



Jim Halverson
arXiv:2408.00082



Daniel A. Roberts et al.
arXiv:2106.10165



Samuel S. Schoenholz et al.
arXiv:1611.01232